

Clinicopathological Features and Outcomes of IgA Nephropathy with Serum Antineutrophil Cytoplasmic Autoantibody Positivity

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Keywords

ANCA-associated systemic vasculitis · Antineutrophil cytoplasmic autoantibodies · IgA nephropathy

Abstract

Introduction: IgA nephropathy (IgAN) with serum antineutrophil cytoplasmic autoantibody (ANCA) positivity is uncommon. This study analyzed the clinicopathologic features and prognosis of IgAN patients with serum ANCA positivity. **Methods:** 2,864 IgAN patients were tested for ANCA by the indirect immunofluorescence assay and chemiluminescence immunoassay. Patients with serum ANCA positivity ($n = 85$) were identified, and their clinical, pathological, and prognostic characteristics were analyzed. They were compared with ANCA-negative IgAN patients ($n = 170$) and ANCA-associated systemic vasculitis (AAV) with renal involvement patients ($n = 85$) selected randomly. **Results:** 2.97% (85/2,864) of IgAN were ANCA positive, and 4 patients were diagnosed as having crescentic IgAN with ANCA positivity. The clinicopathological characteristics of ANCA-positive IgAN patients were comparable to ANCA-negative IgAN patients, but they had higher antinuclear antibody (ANA)-positive rates, lower levels of renal interstitial inflammation, and fewer immune depositions than ANCA-negative IgAN patients. Compared with AAV patients, ANCA-positive IgAN patients were younger and had fewer

extrarenal manifestations, milder renal damage, and more immune complex depositions. The renal outcomes were similar between IgAN patients with and without ANCA positivity. Multivariate Cox analysis revealed that in IgAN patients with ANCA positivity, male, ANA positivity, higher serum creatinine and proteinuria, and more severe renal tubular atrophy/interstitial fibrosis were risk factors for adverse renal outcomes. **Conclusion:** The clinical, pathological features and prognosis of ANCA-positive IgAN patients were similar to those of ANCA-negative IgAN patients except for higher ANA-positive rate, milder renal inflammation, and fewer immune depositions. ANA positivity was an independent risk factor for adverse renal outcomes in ANCA-positive IgAN patients.

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Introduction

IgA nephropathy (IgAN) is a primary glomerular disease with IgA deposition in mesentery, which is the most prevalent primary glomerular disease [1]. Antineutrophil cytoplasmic antibody (ANCA), an autoantibody that activates inflammatory cells and then induces necrotizing small vessel inflammation, is an important

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marker for the diagnosis of ANCA-associated vasculitis (AAV) [2]. There are great differences in clinical manifestations and prognoses between IgAN and AAV, and the coexistence of IgAN and ANCA positivity was very uncommon. There are only case reports [3–19] and 2 small case control studies [20, 21] that reported some of the features of IgAN patients with ANCA positivity. This study explored the clinicopathological and prognostic features of IgAN patients with ANCA positivity in a large consecutive cohort with long time follow-up.

Methods

Study Design and Population

The patients diagnosed with biopsy-proven primary IgAN and received serum ANCA determination in the Department of Nephrology, The First Affiliated Hospital of Sun Yat-sen University from January 1, 2007, to May 31, 2019, were enrolled in this study. Secondary IgAN, such as IgAN caused by Henoch-Schönlein purpura, viral hepatitis, systemic lupus erythematosus et al., and IgAN after renal transplantation were excluded. 170 ANCA-negative biopsy-proven primary IgAN patients and 85 AAV patients who performed renal biopsy in the same time period were selected by simple random sampling as two control groups. AAV patients with renal damage met the Chapel Hill Consensus Conference nomenclature criteria for AAV [22] and also pathologically confirmed renal vasculitis. The diagnosis of IgAN was based on the presence of dominant or codominant staining for IgA in the glomerular mesangium. Crescentic IgAN was defined as IgAN with more than 50% of the total number of glomeruli had large crescent in renal biopsies. P-ANCA/c-ANCA was detected by indirect immunofluorescence (IF), and MPO-ANCA/PR3-ANCA was detected by chemiluminescence immunoassay. If one of the tests was positive, it was clarified as ANCA-positive. The study was approved by the Ethics Committee of the First Affiliated Hospital of Sun Yat-sen University (approval number 2021371). Written informed consent was obtained from each participant.

Data Collection

Clinical and laboratory data including gender, age, the ratio of serum neutrophil to lymphocyte ratio (NLR), hemoglobin, 24 h proteinuria, microscopic hematuria, serum creatinine, urea nitrogen, serum total protein, serum albumin, complement 3 (C3), C-reactive protein, antinuclear antibodies (ANAs), and ANCA at renal biopsy were collected. The extrarenal manifestations of ANCA-positive IgAN patients and AAV patients were evaluated by the Birmingham Vasculitis Activity Score (BVAS) [23]. The estimated glomerular filtration rate (eGFR) was calculated according to the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation [24].

The features of renal biopsy were also analyzed. For IgAN patients, based on the 2016 Oxford IgAN pathological classification standard, MEST-C scores were completed [25]. The IF intensity of IgA, IgG, IgM, C3, and C1q sediments was scored semi-quantitatively on a scale of 0–3+. The deposition of other types of immune complexes other than IgA was defined as a score of + or higher in staining for any kind of immunoglobulin and/or

complement except IgA. The treatment of angiotensin-converting enzyme inhibitor, angiotensin receptor antagonist, glucocorticoids, immunosuppressant, and dialysis was recorded. Immunosuppressants include cyclophosphamide, mycophenolate mofetil, azathioprine, cyclosporine A, tacrolimus, and rituximab.

Follow-Up

The patients were followed from January 1, 2007, until October 30, 2020, or loss to follow-up. The primary endpoint was adverse renal outcomes, defined as a composite of a doubling of serum creatinine or end stage renal disease (ESRD) (eGFR <15 mL/min/1.73 m² for at least 6 months/maintenance dialysis/renal transplantation) or death.

Statistical Analysis

The control ANCA-negative IgAN and AAV groups were randomly selected using SPSS simple random sampling. The results are expressed as a percentage or average $\pm$ SD for normally distributed variables and median (quartile) for non-normally distributed variables. Wilcoxon rank sum test was used to compare the quantitative data of non-normal distribution between the two groups. For the qualitative data, *r* test or Fisher exact probability calculation method (disordered classification variables) or Wilcoxon rank sum test (ordered classification variables) are used for comparison between the two groups. When drawing the Kaplan-Meier survival curve, log-rank test was used to compare whether there was any difference in the survival curve of each group. Univariate and multivariate COX proportional hazard regression analyses were used to examine the relevant factors affecting the long-term prognosis of patients. The multivariate Cox regression model was constructed using eligible covariates that demonstrated significant or near-significant association with outcomes ($p < 0.1$) or for importance of clinical concern. All statistical analyses were carried out using SPSS software version 25.0 and GraphPad Prism software version 8.0. A p value <0.05 was considered to be statistically significant. A p value <0.017 was considered to be statistically significant in post hoc analysis.

Results

Baseline Clinicopathological Characteristics of ANCA-Positive IgAN, ANCA-Negative IgAN, and AAV Patients

The process of patient inclusion is shown in online supplementary material 1 (for all online suppl. material, see <https://doi.org/10.1159/000533982>). The clinicopathological characteristics of ANCA-positive IgAN, ANCA-negative IgAN, and AAV patients are shown in Table 1. The clinical characteristics of ANCA-positive IgAN patients were similar to those of ANCA-negative IgAN patients, but they had higher ANA-positive rate (18.8% vs. 8.2%, $p = 0.012$). Compared with AAV group, the ANCA-positive IgAN group was generally younger (34 [IQR 26–42] vs. 55 [IQR 40–64]) ($p < 0.001$) and had lower BVAS score (34 [IQR 26–42] vs. 55 [IQR 40–64]) ($p < 0.001$) and less extrarenal manifestation

Table 1. Comparison of clinicopathologic parameters and histological data between ANCA-positive IgA nephropathy, ANCA-negative IgA nephropathy, and AAV with renal damage patients

	ANCA-positive IgAN (N = 85)	ANCA-negative IgAN (N = 170)	AAV with renal damage (N = 85)	p value
Gender (male), %	32 (37.6)	73 (41.2)	40 (47.1)	0.460
Age, years	34 (27, 43)	33 (26, 41) ^b	55 (40, 64) ^c	<0.001
BVAS	12 (12, 14)	N/A	16 (14, 19) ^c	<0.001
Extrarenal manifestations, n (%)	30 (35.3)	N/A	75 (88.2) ^c	<0.001
ANCA	p63.5%, c23.5%, p+c 13.0%	N/A	p84.7%, c4.7%, p+c 10.6% ^c	<0.001
Mean arterial blood pressure, mm Hg	93 (87, 106)	96 (87, 106)	97 (88, 106)	0.524
Neutrophil lymphocyte ratio	2.1 (1.5, 3.1)	2.2 (1.5, 3) ^b	3.9 (2.8, 6.9) ^c	<0.001
Hemoglobin, g/L	127.0 (105.5, 135.5)	123.0 (109.8, 136.0) ^b	85.0 (75.0, 100.0) ^c	<0.001
Serum creatinine, μmol/L	104.0 (67.0, 182.0)	89.0 (66.0, 178.5) ^b	452.0 (196.0, 650.0) ^c	<0.001
eGFR, mL/min/1.73 m ²	68.2 (31.8, 108.3)	79.8 (37, 112.3) ^b	10.3 (6.5, 29.0) ^c	<0.001
Blood urea nitrogen, mmol/L	6.2 (4.7, 9.9)	6.1 (4.7, 8.6) ^b	17.6 (10.8, 25.3) ^c	<0.001
Serum total protein, g/L	65.0±8.0	63.5±9.4	64.3±10.0	0.625
Serum albumin, g/L	38.0 (35.0, 41.1)	38.0 (35.0, 41.0) ^b	33.0 (29.0, 37.0) ^c	<0.001
Urinary protein, g/24 h	1.0 (0.4, 1.8)	1.0 (0.4, 2.1) ^b	1.6 (0.9, 3.0) ^c	<0.001
Microscopic hematuria, N (%)	42 (49.4)	108 (63.5)	51 (60.0)	0.095
Gross hematuria, N (%)	16 (18.8)	32 (18.8)	11 (12.9)	0.463
Low complement C3, N (%)	14	32 ^b	28 ^c	0.014
C-reaction protein, mg/L	0.9 (0.8, 1.8)	0.8 (0.8, 2.2) ^b	9.3 (2.3, 34.8) ^c	<0.001
ANA, N (%)	16 (18.8) ^a	14 (8.2) ^b	50 (58.8) ^c	<0.001
Total glomeruli	23 (16, 30)	23 (16, 28)	21 (16, 29)	0.805
Glomeruli with crescents, %	1.8 (0, 7.0)	0 (0, 9.1)	34.5 (21.3, 53.0) ^c	<0.001
Glomeruli with cellular crescents, %	0 (0, 0)	0 (0, 0)	3.4 (0, 11.1) ^c	<0.001
Glomeruli with fibrocellular crescents, %	0 (0, 4)	0 (0, 4.6)	7.1 (0, 17.7) ^c	<0.001
Glomeruli with fibrous crescents, %	0 (0, 0)	0 (0, 0)	18.2 (7.4, 30.6) ^c	<0.001
Global glomerulosclerosis, %	0.2 (0, 0.4)	0.2 (0, 0.5)	0.2 (0.1, 0.5)	0.330
Crescents >50%, N (%)	3 (3.5)	3 (3.5) ^b	26 (30.6) ^c	<0.001
Loop necrosis, N (%)	7 (8.2)	12 (7.1) ^b	32 (37.6) ^c	<0.001
Mesangial hypercellularity, N (%)	54 (63.5)	109 (64.1) ^b	11 (12.9) ^c	<0.001
Endocapillary hypercellularity, N (%)	62 (72.9)	134 (78.8) ^b	82 (96.5) ^c	<0.001
Glomerular segmental sclerosis, N (%)	65 (76.5) ^a	137 (80.6)	33 (38.8) ^c	<0.001
Tubular atrophy/interstitial fibrosis, N (%) (<25%/25–50%/>50%)	56 (65.9)/24 (28.2)/5 (5.9)	108 (63.5)/39 (22.9)/23 (13.5) ^b	32 (37.6)/38 (44.7)/15 (17.6) ^c	<0.001
Interstitial infiltrates, N (%) (<25%/25–50%/>50%)	19 (22.4)/64 (75.3)/2 (2.4) ^a	21 (12.4)/136 (80.0)/13 (7.6)	1 (1.2)/70 (82.4)/14 (16.5) ^c	<0.001
IgA deposition ≥2+	80 (94.1) ^a	168 (98.8) ^b	0 (0) ^c	<0.001
Other deposits ≥1+	66 (77.6) ^a	151 (88.8) ^b	54 (63.5) ^c	<0.001

Normal distribution data were presented as average ± SD and non-normal distribution data as median (Q25, Q75). p value <0.0167 in post hoc analysis among groups. AAV, antineutrophil cytoplasmic autoantibody-associated systemic vasculitis; ANA, antinuclear antibody; ANCA, antineutrophil cytoplasmic antibody; BVAS, Birmingham Vasculitis Activity Score; eGFR, estimated glomerular filtration rate; IgAN, IgA nephropathy. ^aANCA-positive IgAN versus ANCA-negative IgAN. ^bANCA-negative IgAN versus AAV. ^cANCA-positive IgAN versus AAV.

(35.3% vs. 88.2%, $p < 0.001$). Additionally, the patients with ANCA-positive IgAN had a lower level of urea nitrogen ($p < 0.001$), serum creatinine ($p < 0.001$), proteinuria ($p < 0.001$), NLR ($p < 0.001$), C-reactive protein ($p < 0.001$), positive rates of ANA ($p < 0.001$), and percentage of low complement C3 ($p = 0.013$), and higher levels of eGFR ($p < 0.001$), hemoglobin ($p < 0.001$), albumin ($p < 0.001$) compared with the AAV group. The main category of

ANCAs in both groups was p-ANCA or myeloperoxidase ANCA (MPO-ANCA), but the percentage of positive c-ANCA or proteinase 3 ANCA (PR3-ANCA) in the ANCA-positive IgAN group was significantly higher than that in the AAV group ($p < 0.001$).

Compared with the ANCA-negative IgAN group, the ANCA-positive IgAN group showed milder renal interstitial inflammatory cells ($p = 0.012$), a lower

Table 2. Comparison of treatment and prognosis among ANCA-positive IgA nephropathy, ANCA-negative IgA nephropathy and AAV with renal damage patients

Variables	ANCA-positive IgAN (N = 85)	ANCA-negative IgAN (N = 170)	AAV with renal damage (N = 85)	p value
ESRD at onset, N (%)	2 (2.4)	8 (4.7)	9 (10.6)	0.060
ACEI/ARB, N (%)	52 (61.2) ^a	129 (75.9) ^b	17 (20.0) ^c	<0.001
Methylprednisolone pulse, N (%)	15 (17.9)	28 (16.6) ^b	51 (60.0) ^c	<0.001
Oral glucocorticoid, N (%)	40 (47.1)	68 (40.0) ^b	83 (97.6) ^c	<0.001
Immunosuppressant, N (%)	10 (11.8) ^a	5 (2.9) ^b	39 (45.9) ^c	<0.001
Dialysis before biopsy, N (%)	0 (0)	5 (2.9) ^b	18 (10.6) ^c	<0.001
Withdraw from dialysis, % (n/N)	0 (0/0)	20.0 (1/5)	33.3 (6/18)	1.00
Follow-up time, months	41 (25, 65)	48 (27, 72)	18 (6, 38) ^c	<0.001
Adverse renal events, n (%)	13 (15.7)	27 (16.7)	25 (32.9) ^c	0.007
ESRD, % (n/N)	13.3 (11/83)	13.6 (22/162)	25.0 (19/76)	0.058
Creatinine doubled, % (n/N)	10.8 (9/83)	12.3 (20/162)	5.3 (4/76)	0.240
All-cause mortality, N (%)	1 (1.2)	2 (1.2)	7 (9.2) ^c	0.004

p value <0.0167 in post hoc analysis among groups. AAV, antineutrophil cytoplasmic autoantibody-associated systemic vasculitis; ACEI, angiotensin-converting enzyme inhibitor; ANCA, antineutrophil cytoplasmic antibody; ARB, angiotensin receptor blocker; IgAN, IgA nephropathy; ESRD, end stage renal disease. ^aANCA-positive IgAN versus ANCA-negative IgAN. ^bANCA-negative IgAN versus AAV. ^cANCA-positive IgAN versus AAV.

proportion of mesangial IgA deposition ($p = 0.03$), and other types of immune complex deposition ($p = 0.018$) (Table 1). Mesangial hyperplasia ($p < 0.001$) and glomerular segmental sclerosis ($p < 0.001$) were more commonly seen in biopsies of ANCA-positive IgAN patients than those with AAV. Meanwhile, AAV patients showed more capillary endothelial hyperplasia ($p < 0.001$) and crescent formation ($p < 0.001$); at the same time, more severe renal tubular atrophy/interstitial fibrosis ($p < 0.001$) and interstitial inflammatory cell infiltration ($p < 0.001$), when compared with ANCA-positive IgAN patients (Table 1).

Survival Analysis

There were 2, 8, and 9 patients had ESRD at onset and 8, 13, and 12 patients lost to follow-up in the ANCA-positive IgAN, ANCA-negative IgAN, and AAV groups, respectively. The median follow-up time was 41, 48, and 18 months, respectively. Among the ANCA-positive IgAN, ANCA-negative IgAN, and AAV patients, 15.7% (13/85), 16.7% (27/170), and 32.9% (25/85) patients had adverse renal outcomes, respectively. As shown in Table 2, there was no significant difference in the incidence of adverse renal events between ANCA-positive and -negative IgAN patients ($p = 0.841$). But compared with AAV patients, ANCA-positive IgAN patients had a lower incidence of adverse renal events ($p = 0.011$). There was no significant difference in cumulative survival rate for free of

composite adverse renal events between the ANCA-positive and -negative IgAN groups ($p = 0.704$) (Fig. 1a). Cox analysis showed ANCA positivity was not an independent risk factor for adverse events in IgAN patients (online suppl. material 2). Meanwhile, the survival rate of free of renal adverse events in ANCA-positive IgAN patients was higher than that of AAV patients ($p < 0.001$) (Fig. 1b). Further cox regression analysis showed that after adjusting for age and sex, there was still a significant difference in prognosis between ANCA-positive IgAN patients and AAV patients ($p = 0.002$). But after further adjustments of serum creatinine, proteinuria, renal tubular atrophy/interstitial fibrosis, and crescents, there was no significant difference between the two groups ($p = 0.08$), as shown in online supplementary material 2.

Crescentic IgAN with ANCA Positivity

Three female and 1 male patient aged 20–61 years were diagnosed with crescentic IgAN with MPO-ANCA positivity (online suppl. material 3). Patient A had a history of taking propylthiouracil before the onset of kidney disease and was relieved after cyclophosphamide (CTX) and corticosteroid therapy. Patient C also achieved remission after corticosteroid and CTX. Patient B had a history of Sjogren's syndrome (SS), had decreased renal function at onset, and reached ESRD after corticosteroids plus mycophenolate mofetil treatment. Patient D was the only male patient and reached ESRD on admission.

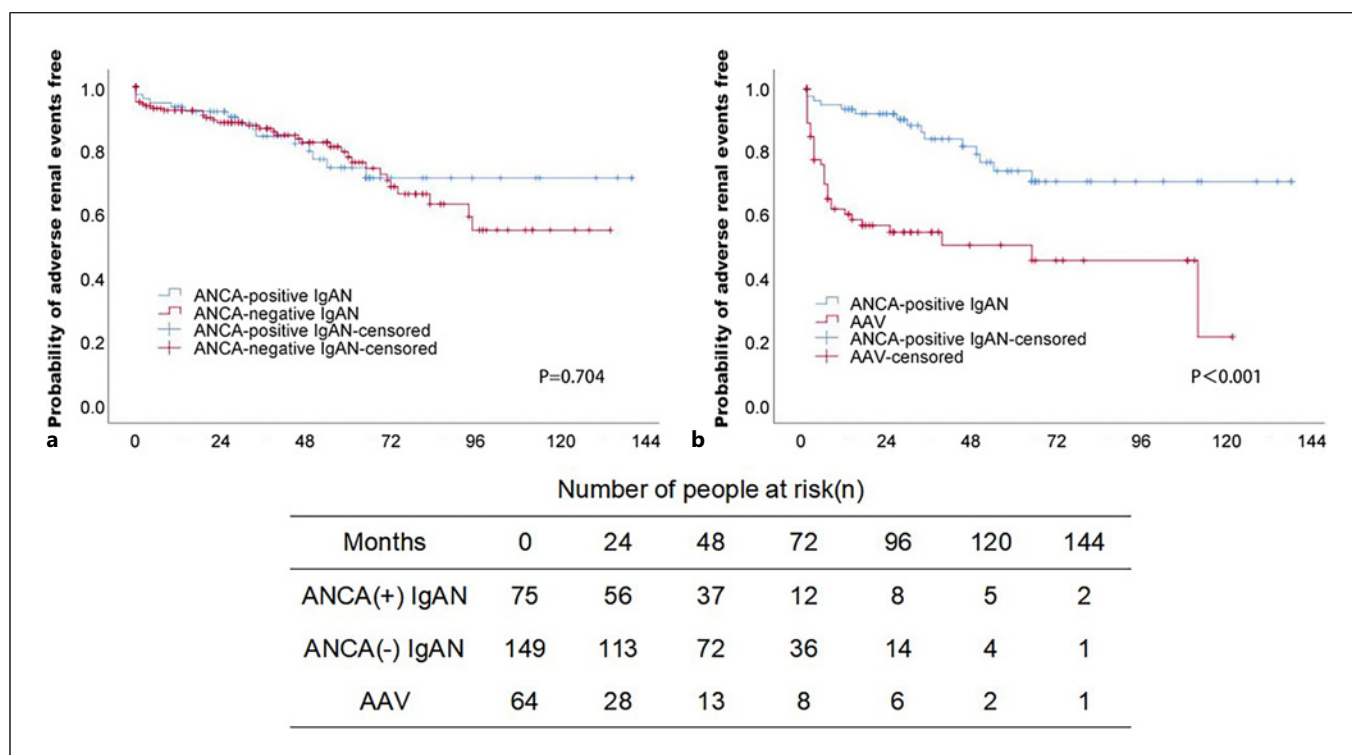


Fig. 1. Kaplan-Meier survival curves analysis of adverse events. **a** ANCA-positive IgAN and ANCA-negative IgAN patients. **b** ANCA-positive IgAN and AAV patients. IgAN, IgA nephropathy; ANCA, anti-neutrophil cytoplasmic antibody; AAV, antineutrophil cytoplasmic auto-antibody-associated systemic vasculitis.

Risk Factors for Adverse Renal Outcomes

Multivariate COX regression analysis (Table 3) showed that male (HR 6.29, 95% CI: 1.26–31.39, $p = 0.025$), higher serum creatinine levels (HR 1.01, 95% CI: 1.00–1.03, $p = 0.006$), and increased proteinuria (HR 1.67, 95% CI: 1.14–2.41, $p = 0.008$), ANA positivity (HR 5.20, 95% CI: 1.06–25.57, $p = 0.043$), and more severe renal tubular atrophy/renal interstitial fibrosis (HR 12.84, 95% CI: 1.78–92.78, $p = 0.011$) were independent risk factors for adverse renal outcomes in ANCA-positive IgAN patients.

Discussion

In the present study, 2.97% (85/2,864) of IgAN were ANCA-positive and 4 patients were diagnosed with crescentic IgAN with ANCA positivity. The clinical and pathological characteristics of ANCA-positive IgAN patients were comparable to ANCA-negative IgAN patients, but they had higher ANA-positive rates, milder renal interstitial inflammation, and fewer immune depositions than ANCA-negative IgAN patients. The renal outcomes were similar between IgAN pa-

Table 3. Multivariable predictors of adverse renal events in ANCA-positive IgA nephropathy patients by multivariate regression analysis

Predictor	<i>p</i> value	Hazard ratio (95% confidence interval)
Age, years	0.98	1.00 (0.96, 1.05)
Sex (male/female)	0.025	6.29 (1.26, 31.39)
Serum creatinine, $\mu\text{mol/L}$	0.006	1.01 (1.00, 1.03)
Proteinuria, g/24 h	0.008	1.67 (1.14, 2.41)
ANA (positive/negative)	0.043	5.20 (1.06, 25.57)
Tubular atrophy/interstitial fibrosis ($\geq 25\%$)	0.011	12.84 (1.78, 92.78)

tients with and without ANCA positivity. Multivariate Cox analysis revealed that in IgAN patients with ANCA positivity, male, ANA positivity, higher serum creatinine and proteinuria, and more severe renal tubular atrophy/interstitial fibrosis were risk factors for adverse renal outcomes.

In previous case series studies, the prevalence of ANCA positivity in IgAN population was about 1.2–2% [17–21], while the positive rate in this study is 2.97%. The

ANCA detection rates of some studies were low or unclear [19–21]; for instance, the ANCA detection rate in the study of Yang et al. [20] was only 48.8%, while the detection rate in our cohort was significantly higher (96.2%). In addition, Bantis et al. [17] only used IF, and Ștefan et al. [19] only used enzyme-linked immunosorbent assay methods to detect ANCA, which may also affect the overall positive rate. Moreover, ANCA was tested in only half of the IgAN patients (50.1%, 1,729/3,454), or the test rate of ANCA was not reported in the previous IgAN cohorts, which may lead to selection bias [20, 21]. Generally speaking, the presence of ANCA in IgAN patients is uncommon. In our study, comparison of ANCA-positive IgAN patients with randomly selected ANCA-negative IgAN patients showed no significance in age, sex, renal function. Moreover, the clinical, pathological features, and prognosis of ANCA-positive IgAN patients were similar to those of ANCA-negative IgAN patients except for higher ANA positive rate, milder renal inflammation, and fewer immune depositions. Well-matched IgAN patients with either ANCA positivity or negativity using propensity score matching for age, sex, and renal functions showed similar results (data not shown). ANCA-positive IgAN patients were more frequently treated with immunosuppressants in our study, and the use of immunosuppressants may influence prognosis. But the efficacy of immunosuppressants in the treatment of IgAN is still controversial in randomized controlled trials [26, 27]. This is only an observational study, so the efficacy of immunosuppressants in the treatment of IgAN with ANCA positivity is uncertain. Bantis et al. [17] studied 8 patients with crescentic IgAN; they showed a more severe clinical picture when compared with ANCA-negative patients. Yang et al. [20] reported that 20 IgAN patients with ANCA positivity had more severe clinical and pathological manifestations than 40 randomly selected IgAN patients with negative ANCA and renal prognosis was relatively better in ANCA-positive crescentic IgAN patients after aggressive immunosuppressive therapy in 6 months. However, Xie et al. [21] studied 35 ANCA-positive IgAN patients and found that there was no significant difference in clinical manifestations and prognostic characteristics between these 35 ANCA-positive IgAN patients and 40 negative IgAN patients after a median follow-up of 9 months. One possible explanation for inconsistencies in the results of previous articles is that epitope specificity determines the pathogenicity and detectability of ANCA [28, 29]. Although ANCA can cause vasculitis and other diseases in clinics, as an autoantibody, it can also exist in healthy individuals and may play a role in maintaining immune

homeostasis [30]. Roth et al. [28] found that MPO autoantibodies from healthy individuals and AAV patients had different epitope specificities. Huang et al. [18] found that not all ANCAs were involved in the pathology of IgAN and suggested that distinguishing pathogenic and non-pathogenic ANCA should be required in patients with IgAN. In this study, because of the higher detection rate and positive probability of ANCA, the inclusion of more non-pathogenic ANCA may make ANCA-positive IgAN patients show mild clinicopathological manifestation and better renal prognosis when compared to AAV patients and were similar to ANCA-negative IgAN patients.

All of the patients (8/8) reported by Bantis et al. [17] and 45% of the ANCA-positive IgAN patients (9/20) reported by Yang et al. [20] showed crescentic glomerulonephritis. Inconsistent with the high percentage of crescentic IgAN reported in the above article, we found only 4/85 (4.7%) were ANCA-positive crescentic IgAN, and there was no significant difference in crescentic proportion between ANCA-negative and ANCA-positive IgAN patients. This may indicate that, among the patients with IgAN, ANCA does not necessarily contribute to the formation of crescents. Usually, the kidney involved by AAV is often characterized by pauci-immune glomerulonephritis, but there is increasing evidence that immune complex deposition can coexist with ANCA-associated glomerulonephritis. Some studies suggested that immune complexes might appear in the early stage of vasculitis and take part in the occurrence of ANCA and the activation of inflammatory reaction, and then initiated to activation of polyclonal B cells and formation of ANCA [3, 31]. In time, the existence of ANCA would aggravate the inflammatory reaction and lead to pauci-immune injury. Immune complexes were degraded by some enzymes released by neutrophils and so on [32, 33]. In this study, the overall deposition intensity of mesangial IgA and other immune complexes in ANCA-positive IgAN patients was relatively weaker than that in ANCA-negative IgAN patients, which may be related to the degradation of immune complexes because of ANCA. More studies are needed to clarify to mechanism of ANCA in IgAN. Existing studies had shown that there were differences in regional population distribution and clinical characteristics among AAV patients with different ANCA types [34–37]. Compared with MPO-ANCA-positive patients, PR3-ANCA-positive patients were younger, had fewer kidney involvements, and had more extrarenal manifestations [34, 35]. In this study, the positive rates of c-ANCA and/or PR3-ANCA in ANCA-positive IgAN patients were significantly higher than those in AAV patients. The population in our study is all Chinese and the results may not be generalizable to Caucasians or

other patients. In ANCA-positive IgAN, 22/85 (25.8%) are positive only for pANCA or cANCA. 63/85 (74.1%) patients are positive for MPO-ANCA or PR3-ANCA. It is difficult to exclude AAVS for some ANCA-positive IgAN patients, and it is not easy to definitely differentiate between AAVS with mesangial IgA deposits and IgAN with ANCA positivity in clinical practice. The results of multivariate COX proportional hazard regression analysis showed that male, higher serum creatinine and proteinuria, more severe renal tubulointerstitial fibrosis, and ANA positivity were independent risk factors for adverse renal outcomes in ANCA-positive IgAN patients. Male, higher serum creatinine and proteinuria, more severe renal tubulointerstitial fibrosis are common factors affecting the prognosis of IgAN patients [38, 39], but the role of ANA was rarely reported in the past. As a kind of autoantibodies, ANA frequently appears in AAV, and its positive rate is significantly higher than that of other types of autoantibodies [40]. However, although IgAN is considered to be an immune-mediated disease, according to previous studies, it is not statistically related to the incidence of elevated ANA [41, 42]. In this study, the positive rate of ANA in AAV patients was high (58.8%). In IgAN patients, the ANA-positive rate of ANCA-positive patients was also significantly higher than that of ANCA-negative patients (19% vs. 8.2%). Yang et al. [20] reported that the positive rate of ANA in ANCA-positive IgAN patients was significantly higher than that in ANCA-negative IgAN patients, which was consistent with the results of this paper. Some cases suggest that when both ANA and ANCA are positive in AAV, there may be more severe clinical and histological manifestations [16, 43–46]. For example, Zhao et al. [43] found that damage was more severe, histopathological changes were more chronic, and the possibility of reaching ESRD was higher in ANA-positive AAV patients renal than in those who were ANA-negative, suggesting that ANA was of great significance to the prognosis of AAV patients. Moreover, data about other lupus autoantibodies were not collected in this study. There may be some false ANA positivity in ANCA-positive patients. However, due to the small number of endpoint events in this study, the role of ANA in IgAN patients with ANCA positivity should be validated in a larger cohort, and the mechanism needs to be explored further.

Our study has several limitations as it is retrospective and single-centered and the median length of follow-up was relatively short, which requires further multicenter prospective cohort studies with longer follow-up time to validate the findings. Moreover, we only studied the adverse renal outcomes but did not evaluate the response to treatment and recurrence of the diseases, which influenced the analysis of the prognosis of the patients. The results of

the two test methods, IF and chemiluminescence immunoassay for ANCA, are inconsistent in 22/85 (25.8%) of ANCA-positive IgAN patients. Because of the limited cases, further subgroup analyses were not conducted. Further studies are needed to compare different methods for ANCA test in IgAN patients and the clinical significance. The lack of data about the dynamic changes of ANCA during the follow-up period was another limitation.

In conclusion, the clinical, pathological, and prognostic features of ANCA-positive IgAN patients were similar to those of ANCA-negative IgAN patients except for higher ANA positive rate, milder renal inflammation, and immune deposition. ANA positivity was an independent risk factor for adverse renal outcomes in ANCA-positive IgAN patients. Further investigations are needed to study the role of ANCA and ANA in the development and progression of IgAN.

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Statement of Ethics

The study was approved by the Ethics Committee of the First Affiliated Hospital of Sun Yat-sen University, approval number 2021371. Written informed consent was obtained from each participant. We have received written consent for the publication of these details.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

Wenchao Li, Xi Xia, and Wei Chen conceived the study; Wenchao Li, Ruifan Chen, and Xi Xia contributed to acquisition of data; Wenchao Li, Ruifan Chen, Xi Xia, Wei Chen, and Fengxian

Huang contributed to interpretation of data; Xi Xia drafted the manuscript; Wenchao Li, Ruifan Chen, Wei Chen, and Fengxian Huang revised and completed the manuscript. All authors approved the final version to be published and agreed to be accountable for all aspects of the work.

Data Availability Statement

All data generated or analyzed during this study are included in this article and its online supplementary material files. Further inquiries can be directed to the corresponding author.

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